

Project Execution Schedule

Behind the Seens — The Autism & Emotional AI Platform | Technical Execution & Engineering Timeline (Sep 2026 – Feb 2028)

1. WORK BREAKDOWN STRUCTURE (WBS)

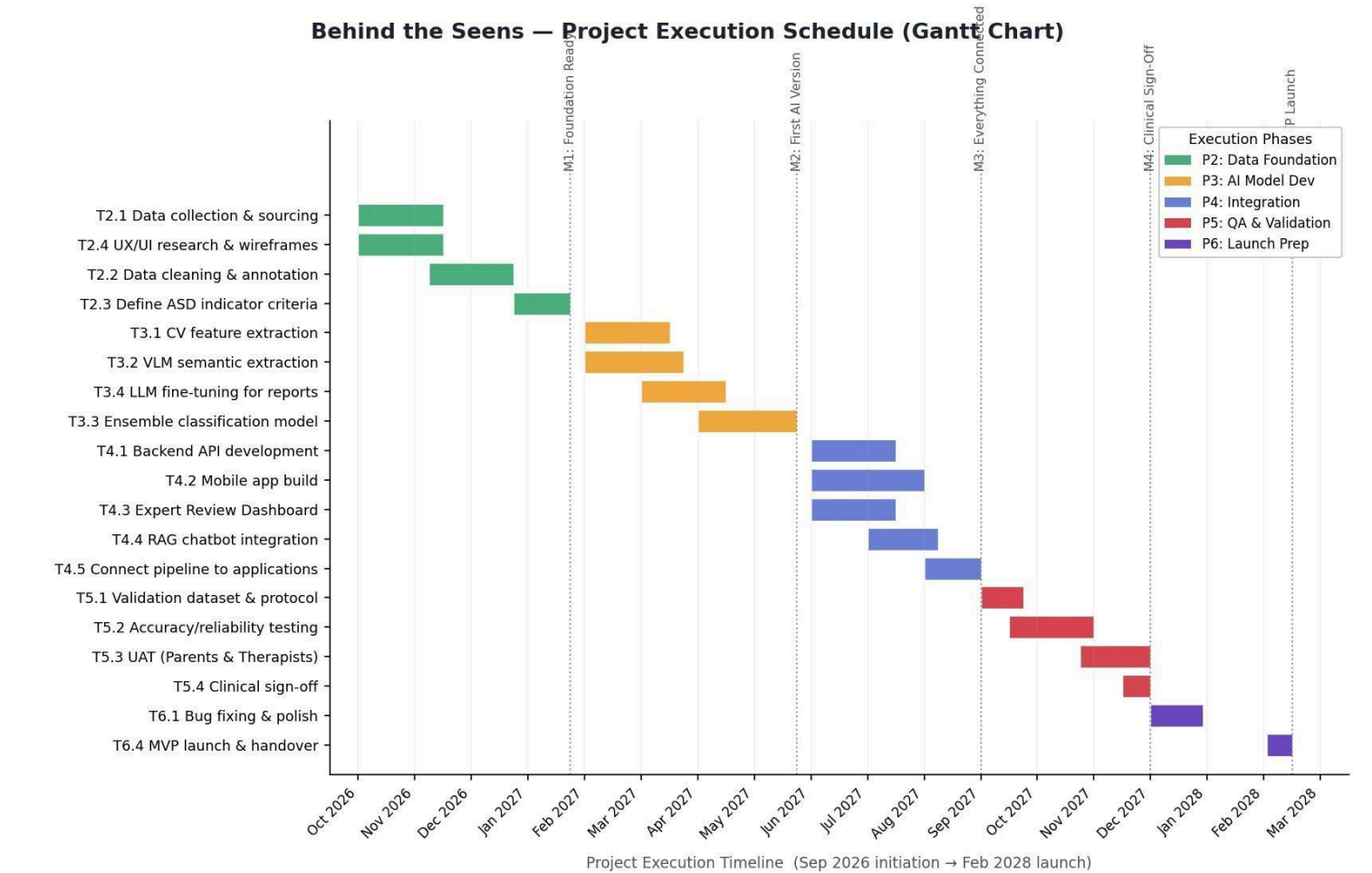
Covers the active engineering, data, and validation phases (P2–P6). Initiation & administrative planning (P1) is covered separately in the Project Charter and is not repeated here, to keep this document focused on the technical critical path.

TASK ID & NAME	PRIMARY OWNER	DURATION
P2 — Data Foundation		
T2.1 Data collection & sourcing	Data Engineer	6 weeks
T2.4 UX/UI research & wireframes	UX/UI Designer	6 weeks
T2.2 Data cleaning & annotation	Data Engineer	5 weeks
T2.3 Define ASD indicator criteria	Clinical Advisory Committee & AI Engineers	4 weeks
P3 — AI Model Development		
T3.1 CV feature extraction	AI Engineers	6 weeks
T3.2 VLM semantic extraction	AI Engineers	7 weeks
T3.4 LLM fine-tuning for reports	AI Engineers	6 weeks
T3.3 Ensemble classification model	AI Engineers	7 weeks
P4 — Integration		
T4.1 Backend API development	Full-stack Developer	6 weeks
T4.2 Mobile app build	Full-stack Developer	8 weeks
T4.3 Expert Review Dashboard	Full-stack Developer & UX/UI Designer	6 weeks
T4.4 RAG chatbot integration	AI Engineers & Full-stack Developer	5 weeks
T4.5 Connect pipeline to applications	Full-stack Developer	4 weeks
P5 — QA & Validation		
T5.1 Validation dataset & protocol	Clinical Advisory Committee	3 weeks
T5.2 Accuracy/reliability testing	QA Engineer	6 weeks (starts week 3, 1-week overlap with T5.1)
T5.3 UAT (Parents & Therapists)	QA Engineer & Clinical Advisory Committee	5 weeks (starts week 7, 2-week overlap with T5.2)
T5.4 Clinical sign-off	Clinical Advisory Committee	2 weeks (starts week 11, overlaps final UAT results)
P6 — Launch Preparation		
T6.1 Bug fixing & polish	Execution Team	4 weeks
T6.4 MVP launch & handover	Execution Team	2 weeks

2. FIX NOTE — P5 SCHEDULE COMPRESSION

The original sequential P5 plan (3+6+5+2 = 16 weeks) did not fit inside the 13-week window between Month 13 (Everything Connected) and Month 16 (Clinical Sign-Off). This version applies **schedule fast-tracking** — a standard PM technique for compressing a critical path by starting a task before its predecessor fully finishes, where the partial output already supports it (e.g. QA can begin accuracy testing once the validation protocol is drafted, without waiting for the protocol document to be formally finalized). With the overlaps above, P5 now fits in 13 weeks and still lands exactly on the Month 16 gate.

3. EXECUTION GANTT CHART



4. EXECUTION MILESTONES & QUALITY GATES

M1 – Jan 2027	Foundation Ready — data pipelines active; ASD baseline criteria locked by the Clinical Advisory Committee. Hard gate before modeling can begin.
M2 – Jun 2027	First AI Version — ensemble and fine-tuned models complete training and enter backend API integration.
M3 – Sep 2027	Everything Connected — end-to-end pipeline mapped (Mobile App → AI Pipeline → Expert Dashboard). Ready for validation dataset tests.
M4 – Dec 2027	Clinical Sign-Off — the most critical execution gate. The system must hit the ≥ 80% accuracy threshold and pass UAT to proceed to launch prep.
M5 – Feb 2028	MVP Launch — code freeze, final performance polish, and technical handover to pilot users.

5. TECHNICAL DEPENDENCIES & PARALLEL WORKSTREAMS

- **Data vs. Interface:** Mobile UX/UI wireframing (T2.4) runs in parallel with backend data gathering (T2.1).
- **Dual-Model Training:** Computer Vision extraction (T3.1) and VLM semantic extraction (T3.2) run synchronously, feeding into the Ensemble classification (T3.3) downstream.
- **Integration Alignment:** Backend API infrastructure (T4.1) and Mobile App (T4.2) development launch concurrently the moment M2 is cleared, merging at T4.5.
- **QA Fast-Tracking:** Validation protocol (T5.1), accuracy testing (T5.2), and UAT (T5.3) run with deliberate 1–2 week overlaps so the full QA cycle fits inside the Month 13–16 window (see Section 2).